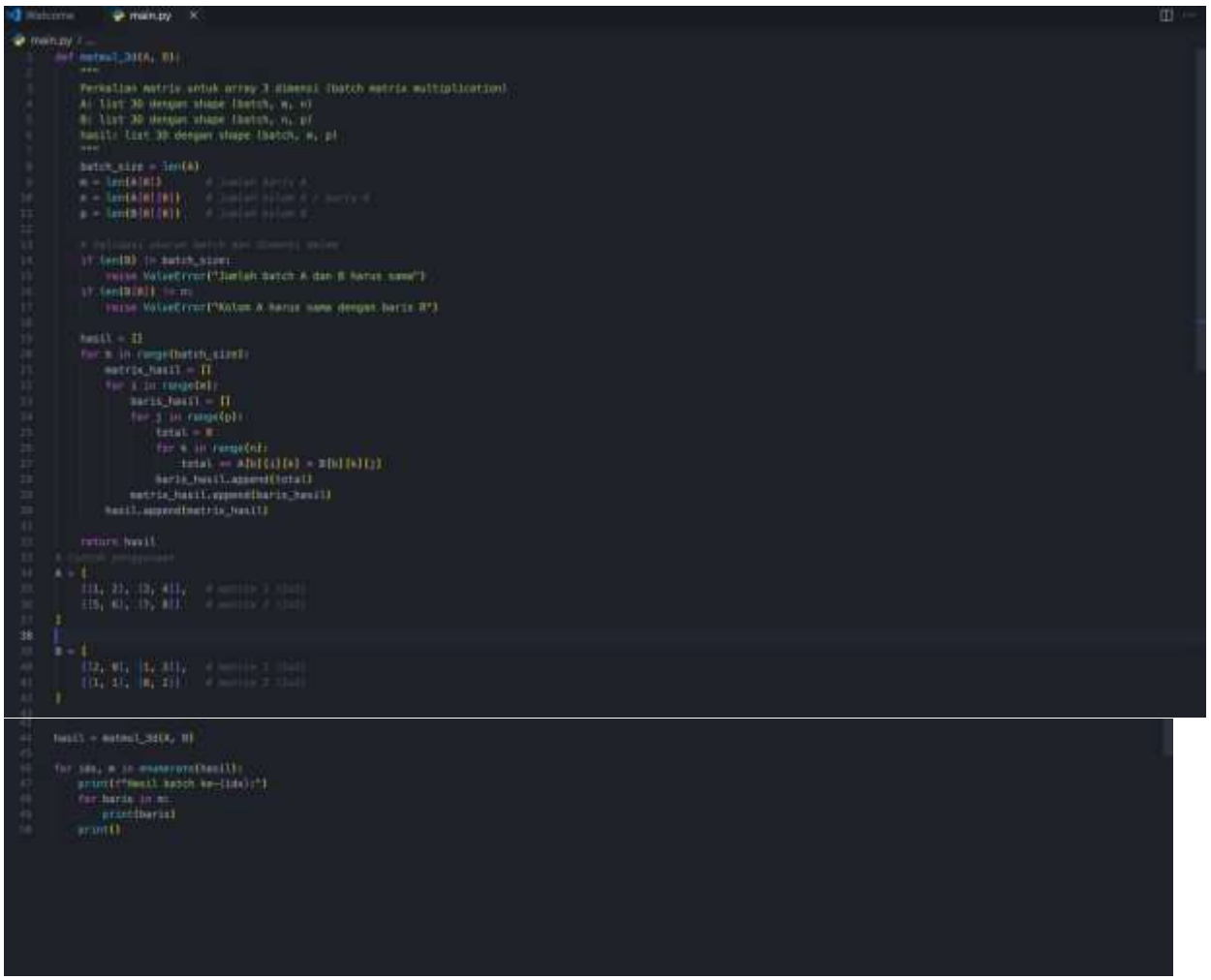
TUGAS PENGOLAHAN CITRA DIGITAL

DEWA RADITYA SASKARA
F1G124061

1. Kode Program



```
def matmul_3d(A, B):
    """
    Perkalian matriks untuk array 3 dimensi (batch matrix multiplication)
    A: list 3D dengan shape (batch, n, o)
    B: list 3D dengan shape (batch, o, p)
    hasil: list 3D dengan shape (batch, n, p)
    """
    batch_size = len(A)
    n = len(A[0])      # jumlah baris A
    o = len(A[0][0])   # jumlah kolom A / baris B
    p = len(B[0][0])   # jumlah kolom B

    # Validasi: ukuran batch dan dimensi cocok
    if len(B) != batch_size:
        raise ValueError("Jumlah batch A dan B harus sama")
    if len(B[0]) != o:
        raise ValueError("Kolom A harus sama dengan baris B")

    hasil = []
    for b in range(batch_size):
        matrix_hasil = []
        for i in range(n):
            baris_hasil = []
            for j in range(p):
                total = 0
                for k in range(o):
                    total += A[b][i][k] * B[b][k][j]
                baris_hasil.append(total)
            matrix_hasil.append(baris_hasil)
        hasil.append(matrix_hasil)

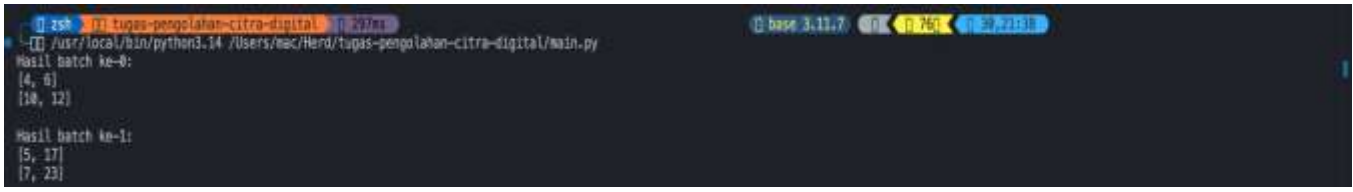
    return hasil
# Contoh penggunaan
A = [
    [[1, 2], [3, 4]],   # matriks 1 (2x2)
    [[5, 6], [7, 8]]    # matriks 2 (2x2)
]

B = [
    [[2, 0], [1, 3]],   # matriks 1 (2x2)
    [[1, 1], [0, 2]]    # matriks 2 (2x2)
]

hasil = matmul_3d(A, B)

for idx, m in enumerate(hasil):
    print(f"Hasil batch ke-{idx}:")
    for baris in m:
        print(baris)
    print()
```

2. Output


zsh
tugas-pengolahan-citra-digital
/usr/local/bin/python3.14 /Users/mac/Herd/tugas-pengolahan-citra-digital/main.py
Hasil batch ke-0:
[4, 6]
[10, 12]
Hasil batch ke-1:
[5, 17]
[7, 23]